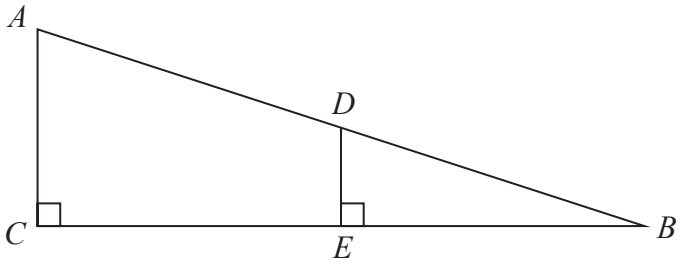
10 The diagram shows two right-angled triangles *ACB* and *DEB*.



AD = 9 cm
DE = 2 cm
DB = 6 cm

Calculate the length of *CB*.
Give your answer correct to 2 decimal places.

Small triangle: EB = $\sqrt{(6^2 - 2^2)} = \sqrt{32}$

Whole hypotenuse AB = AD + DB = 9 + 6 = 15

Similar-triangle scale factor = AB / DB = 15/6 = 2.5

CB = $2.5\sqrt{32} = 10\sqrt{2} = 14.142...$

CB = 14.14 cm (2 d.p.)

1 mark
scale factor 2.5

1 mark
EB = √32

1 mark
CB = 2.5 EB

1 mark
14.14 cm

14.14 cm

(Total for Question 10 is 4 marks)

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